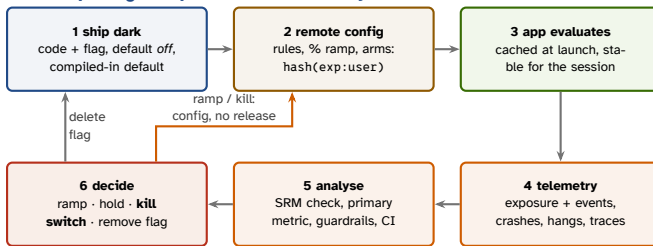
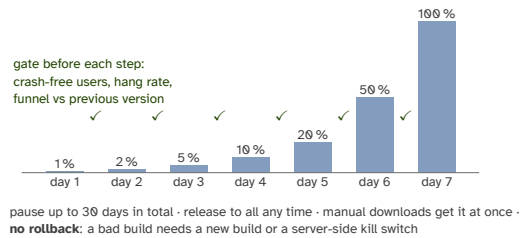


In one line: Ship code **dark** behind a **flag**, turn it on for a **hashed, random slice** of users, **log who was exposed**, compare their metrics with control while **guardrails** (crashes, hangs, revenue) watch for harm, then ramp, hold or **kill** — without a new binary. On mobile this matters twice: a shipped build lives for months and the store rollout cannot be rolled back.

The loop: flag + experiment + telemetry



App Store phased release (automatic updates)



Feature flags

kind	purpose	lifetime · decided by
release	hide unfinished work, trunk-based dev	days-weeks · per build/env
experiment	A/B arms	weeks · per user, hashed
ops	kill switch, degrade under load	short or permanent · ops, at runtime
permission	premium, beta, internal	long · per user/entitlement

- Evaluation:** compiled-in defaults (first launch, offline) → last-known-good cache on disk → fetch at launch with a short timeout, applied at a *safe point* (next launch / before the screen builds) — never flip UI mid-session. Experiments read once per session.
- Kill switch:** server-side off for a risky path without a release; test the off path too. Add a minimum-version / force-update switch before you need it.
- Local overrides:** debug menu, launch arguments (–newCheckout YES lands in UserDefaults' argument domain), UI tests set flags explicitly — never depend on the live server.
- Flag debt:** owner + expiry per flag, delete at 100%, lint for stale keys. Mobile twist: old app versions still read the key — keep a server default until they are gone. n flags = 2^n paths.

A/B testing

- Randomisation unit** = analysis unit: user id (stable across devices) > install id (resets on reinstall) > session.
- Bucketing:** $\text{hash}(\text{experimentKey} + ":" + \text{userId}) \bmod 10000$; arm = range. Deterministic, sticky, stateless; the salt makes experiments independent; ramping by widening a range keeps users in place.
- Exposure logging:** log `experiment_exposure` when the user *reaches* the changed code, not at assignment; analyse exposed users only (otherwise the effect is diluted).
- SRM** (sample-ratio mismatch): 50/50 planned, 50.8/49.2 at large $n \rightarrow \chi^2$ fails → assignment or logging bug; results void.
- Sample size** ($\alpha = 0.05$ two-sided, power 80%): $n_{\text{arm}} \approx 16 \sigma^2 / \delta^2$; conversion $\sigma^2 = p(1-p)$. 10% → 11%: $16 \cdot 0.09 / 0.01^2 \approx 14\,400$ per arm. Smaller effect → quadratically more users.
- Peeking:** checking daily and stopping at the first $p < 0.05$ inflates false positives far above 5%. Fix the horizon (full weeks), or use a sequential test built for it.
- Guardrails:** crash-free users, latency, revenue, uninstalls must not regress even when the primary metric wins. Many metrics → some “win” by chance.

Analytics event design

- Name** `object_action`, past tense, snake_case: `checkout_started`, `item_added`. A **tracking plan** (schema registry) types every property; validate in debug and CI.
- Envelope:** `event_id` (UUID, server dedupe), client time + server receive time (skewed clocks), anonymous/user id, session id, app version + build, OS, device, active experiment arms.
- No PII:** no email, name, free text or precise location; pseudonymous ids; honour consent (ATT for cross-app tracking). The privacy manifest and App Store privacy label must match what you send.
- Transport:** buffer → persisted queue → batch flush by count, time or app-backgrounding; gzip; retry with backoff; cap the queue (drop oldest, count drops); `event_id` makes retries idempotent.

Example — flags, bucketing, exposure

```
import CryptoKit
struct Flags: Codable { var newCheckout = false }
final class FlagStore { // read once per session
    private(set) var current: Flags
    init(cached: Data?) { // disk cache, else defaults
        current = cached.flatMap { try? JSONDecoder()
            .decode(Flags.self, from: $0) } ?? Flags() }
    func refresh() async { /* fetch; save for NEXT launch */ }
}
func bucket(_ exp: String, _ user: String) -> Int { // 0..9999
    let d = SHA256.hash(data: Data("\(exp):\(\user)".utf8))
    return Int(d.prefix(4).reduce(UInt32(0)) { $0 << 8 | UInt32($1) }
        % 10_000) // NOT hashValue
}
let arm = bucket("checkout_v2", uid) < 5_000 ? "control" : "test"
analytics.track("experiment_exposure", // at the screen
    ["experiment": "checkout_v2", "arm": arm])
```

Mobile observability

- Crash-free users** = users with no crash / active users; **crash-free sessions** = sessions with no crash / sessions. Users hides frequency, sessions hides breadth — watch both, per version.
- Symbolication: upload dSYMs per build from CI, or stacks stay hex. **Hangs** (iOS hang rate, s/hour) ~ Android **ANR**. **Traces:** spans for cold start, screen load, API calls, trace id propagated to the backend. **Breadcrumbs:** last N screens, requests, logs attached to each crash; log non-fatal errors too. (Crashlytics, Sentry, Datadog: examples.)
- Release gate:** compare the new version with the previous one over the same days and a minimum sample; pause the phased release on a regression; separate *binary* rollout from *feature* rollout.

Traps

- ! Bucketing with Swift `hashValue` — seeded per process: arms change every launch.
- ! Logging exposure at launch for every user — dilutes the effect.
- ! “Phased release lets me roll back” — it only pauses; ship a fix or flip a flag.
- ! Stopping when p dips under 0.05 (peeking); ignoring SRM.
- ! Randomising a signed-in feature by device: one user sees both arms on iPhone and iPad — pick the unit you analyse.

Remember

Default off, hash to bucket, log exposure, guard the rails, ramp in steps, delete the flag.